Topic 1: HKEX & CME Regulations

Guidelines:

- Give an overview on HKEX and CME.
- Introduce OTC Clear margin, and its margin methodology. Make comparison between HKEX and CME.
- Compare PRiME (HKEX) and SPAN (CME).
- Compare SPAN and SPAN 2 framework.
- Illustrate PRiME or SPAN methodology with application on products.

Topic 2: Callable Bull/Bear Contracts (CBBC)

Guidelines:

- Give an introduction on CBBC, explain the features of it.
- Introduce client margin methodologies on CBBC.
- Analyze value-at-risk and expected shortfall (if possible) of the longing/shorting the product by simulations.
- Margin account illustration. Compare VaR estimation and SPAN.

Topic 3: Index Futures

Guidelines:

- Give an introduction on index futures.
- Introduce related risk measurement methodologies on trading futures.
- Analyze value-at-risk and expected shortfall (if possible) of the longing/shorting the product by simulations.
- Margin account illustration.

Topic 4: Cryptocurrency Futures

Guidelines:

- Give an introduction on cryptocurrencies and digital assets.
- Introduce related risk measurement methodologies on cryptocurrencies.
- Model estimation based on GARCH or another model you find appropriate.
- Analyze value-at-risk and expected shortfall (if possible) of the longing/shorting the product using filtered historical simulations.
- Benchmark with results based on Black-Scholes model
- Backtest.

Topic 5: Commodity Options

Guidelines:

- Give an introduction on commodity options.
- Introduce related risk measurement methodologies on commodity.
- Model estimation based on Merton's Jump-diffusion model or another model you find appropriate.
- Analyze value-at-risk and expected shortfall (if possible) of the longing/shorting the product by simulations.
- Benchmark with results based on Black-Scholes model
- Backtest.

Topic 6: VIX Options

Guidelines:

- Give an introduction on VIX options and the VIX itself, explaining the concept of volatility index.
- Introduce related risk measurement methodologies on VIX options.
- Model estimation based on Heston's Stochastic Volatility model or another model you find appropriate.
- Analyze value-at-risk and expected shortfall (if possible) of longing/shorting the product by simulations.
- Backtest.

Topic 7: Foreign Exchange Derivatives

Guidelines:

- Give an introduction on FX option.
- Introduce related risk measurement methodologies on FX derivatives.
- Calibrate a GARCH model for the underlying exchange rate.
- Analyze value-at-risk and expected shortfall (if possible) of the longing/shorting the product by simulations.
- Benchmark with the results based on Black-Scholes.
- Backtest.

Topic 8: Interest Rate Range Accrual Notes

Guidelines:

- Give an introduction on interest rate range accrual notes.
- Introduce related risk measurement methodologies on interest rate range accrual notes.
- Calibrate the Hull-White model for the short rate.
- Analyze value-at-risk and expected shortfall (if possible) of the longing/shorting the product by simulations.
- Apply direction simulation approach and trinomial tree approach in simulation and compare the results.
- Backtest.

Topic 9: Interest Rate Swaps

Guidelines:

- Give an introduction on selected interest rate swaps.
- Introduce related risk measurement methodologies on interest rate swaps.
- Introduce the history of London Interbank Offered Rate (LIBOR) market and the new Secured Overnight Financing Rate (SOFR) market.
- Model the yield curve and do calibration.
- Analyze value-at-risk and expected shortfall (if possible) of the longing/shorting the product by simulations.
- Backtest.

Topic 10: Simultaneously Long-short Strategy

Guidelines:

- Give an introduction on simultaneously long-short trading (SLS) strategy.
- Construct portfolio consists of some stocks from S&P500 based on SLS.
- Assume the assets follows GARCH model: calibration of parameters will be needed. (could be other stochastic volatility model)
- Analysis value-at-risk and expected shortfall (if possible) of proposed portfolio by simulations.
- Benchmark the proposed portfolio with Black-Scholes model based portfolios, compare VaR and ES.
- Analysis the result based on empirical data.

Useful links:

HKEX

http://www.hkex.com.hk/eng/index.htm

HKEX OTC

https://www.hkex.com.hk/Services/Clearing/OTC-Clear/Overview/Clearing-Services?sc_lang=en

OTC Clearing Rules and Clearing Procedures

https://www.hkex.com.hk/Services/Rules-and-Forms-and-Fees/Rules/OTC-Clear/Rules?sc_lang=en

1. HIBOR Future

https://www.hkex.com.hk/Market-Data/Futures-and-Options-Prices/Interest-Rate/Three-Month-HIBOR-Futures?sc_lang=en#&product=HB3

2. Stock Option

https://www.hkex.com.hk/Products/Listed-Derivatives/Single-Stock/Stock-Options?sc_lang=en

3. Interest Rate Swap (OTC)

https://www.hkex.com.hk/Products/OTC-Derivatives/Interest-Rate-Swaps?sc_lang=en

4. Non-deliverable Forward (OTC)

https://www.hkex.com.hk/Products/OTC-Derivatives/Non-Deliverable-Forwards?sc_lang=en

5. Callable Bull/Bear Contract

https://www.hkex.com.hk/Global/Exchange/FAQ/Products/Securities/CBBC?sc_lang=en

6. FX Futures and Options

https://www.hkex.com.hk/Market-Data/Futures-and-Options-Prices/Foreign-Exchange?sc_lang=en

7. Commodities Products

https://www.hkex.com.hk/Products/Listed-Derivatives/Commodities/USD-Gold?sc_lang=en#&product=GDU

CME

http://www.cmegroup.com/

CME OTC

http://www.cmegroup.com/trading/otc/

CME Risk Management Overview

http://www.cmegroup.com/clearing/risk-management/span-overview.html

1. Commodity Futures

http://www.cmegroup.com/markets/metal/metals.html

2. Cryptocurrencies

https://www.cmegroup.com/markets/cryptocurrencies.html

3. Options

http://www.cmegroup.com/markets/options.html

4. FX Derivative

https://www.cmegroup.com/trading/fx/otc.html

5. VIX Options

https://www.cboe.com/tradable_products/vix/vix_options/

6. Interest Rates Swap

https://www.cmegroup.com/markets/interest-rates.html

http://www.cmegroup.com/education/files/cleared-otc-irs-margin-methodology.pdf